

1. Chargement des données

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings

warnings.filterwarnings("ignore")

dataset = pd.read_csv("dataset.csv")
```

2. Analyse exploratoire (EDA)

```
dataset.head(10)
```

	nom	prenom	sexe	date_naissance	lieu_naissance	\
0	Dupont	Alice	F	1985-03-12	Tours	
1	Martin	Bob	M	1978-11-02	Paris	
2	Durand	Charlie	M	1990-07-25	Lyon	
3	Lefevre	Emma	F	1982-01-17	Bordeaux	
4	Moreau	David	M	1975-09-09	Lille	
5	NaN	NaN	M	1988-06-10	Marseille	
6	Dubois	NaN	F	NaN	Nice	
7	NaN	NaN	F	NaN	NaN	
8	Petit	Lucas	M	1995-04-04	Nantes	
9	Robert	Chloe	F	1989-12-30	Strasbourg	

	adresse	commune	pays	nir
organisme \				
0	12 rue Victor Hugo	Tours	France	2.850312e+14
CPAM				
1	45 avenue République	Paris	France	1.781103e+14
MSA				
2	8 rue Nationale	Lyon	France	1.900725e+14
CPAM				
3	22 rue Sainte Catherine	Bordeaux	France	2.820117e+14
RSI				
4	3 rue Faidherbe	Lille	France	1.750909e+14
CPAM				
5	NaN	Marseille	France	NaN
CPAM				
6	15 rue Massena	Nice	France	NaN
MSA				
7	NaN	NaN	France	NaN
NaN				

```

8          9 rue Crébillon      Nantes  France  1.950404e+14
CPAM
9  5 rue des Grandes Arcades  Strasbourg  France  2.891230e+14
RSI

    employeur      bin      nic  date_debut  date_fin      regime
identifie
0   Carrefour  12345.0  678.0  2010-01-01  2015-06-30      General
1
1       Total  54321.0  111.0  2005-03-01  2018-12-31      Agricole
1
2       Amazon  67890.0  222.0  2016-05-01  2020-10-15      General
1
3  Indépendant  22222.0  333.0  2012-09-01      NaN  Independant
1
4       Auchan  33333.0  444.0  2000-01-01  2010-12-31      General
1
5          NaN  12345.0  555.0  2014-02-01      NaN      General
0
6          NaN  54321.0  666.0      NaN      NaN      Agricole
0
7          NaN      NaN      NaN      NaN      NaN      NaN
0
8       Leclerc  11111.0  777.0  2018-06-01      NaN      General
1
9  Freelance  22222.0  888.0  2015-01-01      NaN  Independant
1

```

```

# (nombre de lignes, nombre de colonnes)
dataset.shape

```

```

(47, 17)

```

```

# des informations sur le jeu de données
dataset.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 47 entries, 0 to 46
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   nom                    32 non-null    object
1   prenom                 30 non-null    object
2   sexe                   47 non-null    object
3   date_naissance         34 non-null    object
4   lieu_naissance         34 non-null    object
5   adresse                34 non-null    object
6   commune                36 non-null    object
7   pays                   47 non-null    object
8   nir                    30 non-null    float64

```

```

9  organisme      36 non-null    object
10 employeur      33 non-null    object
11 bin            36 non-null    float64
12 nic            36 non-null    float64
13 date_debut     35 non-null    object
14 date_fin       14 non-null    object
15 regime         39 non-null    object
16 identifie      47 non-null    int64

```

```
dtypes: float64(3), int64(1), object(13)
```

```
memory usage: 6.4+ KB
```

```
# lister les valeurs catégorielles
```

```
dataset.select_dtypes(include="object").columns
```

```

Index(['nom', 'prenom', 'sexe', 'date_naissance', 'lieu_naissance',
      'adresse',
      'commune', 'pays', 'organisme', 'employeur', 'date_debut',
      'date_fin',
      'regime'],
      dtype='object')

```

```
# la taille des valeurs catégorielles
```

```
len(dataset.select_dtypes(include="object").columns)
```

```
13
```

```
# lister les valeurs numériques
```

```
dataset.select_dtypes(include="number").columns
```

```
Index(['nir', 'bin', 'nic', 'identifie'], dtype='object')
```

```
# la taille des valeurs numériques
```

```
len(dataset.select_dtypes(include="number").columns)
```

```
4
```

```
# les statistiques
```

```
dataset.describe(include="all")
```

	nom	prenom	sexe	date_naissance	lieu_naissance
adresse \					
count	32	30	47	34	34
unique	32	25	2	34	32
top	Dupont	Lucas	M	1985-03-12	Lyon
Victor Hugo					12 rue
freq	1	3	24	1	2
1					
mean	NaN	NaN	NaN	NaN	NaN
NaN					

std	NaN	NaN	NaN	NaN	NaN	NaN
NaN						
min	NaN	NaN	NaN	NaN	NaN	NaN
NaN						
25%	NaN	NaN	NaN	NaN	NaN	NaN
NaN						
50%	NaN	NaN	NaN	NaN	NaN	NaN
NaN						
75%	NaN	NaN	NaN	NaN	NaN	NaN
NaN						
max	NaN	NaN	NaN	NaN	NaN	NaN
NaN						
	commune	pays	nir	organisme	employeur	bin
\						
count	36	47	3.000000e+01	36	33	36.000000
unique	32	1	NaN	3	18	NaN
top	Lyon	France	NaN	CPAM	Amazon	NaN
freq	3	47	NaN	22	4	NaN
mean	NaN	NaN	2.346327e+14	NaN	NaN	52223.805556
std	NaN	NaN	5.368592e+13	NaN	NaN	28767.576551
min	NaN	NaN	1.750909e+14	NaN	NaN	11111.000000
25%	NaN	NaN	1.853158e+14	NaN	NaN	24977.500000
50%	NaN	NaN	1.955367e+14	NaN	NaN	54321.000000
75%	NaN	NaN	2.905999e+14	NaN	NaN	75333.000000
max	NaN	NaN	2.980404e+14	NaN	NaN	99999.000000
	nic	date_debut	date_fin	regime	identifie	
count	36.000000	35	14	39	47.000000	
unique	NaN	33	14	3	NaN	
top	NaN	2020-01-01	2015-06-30	General	NaN	
freq	NaN	2	1	27	NaN	
mean	483.166667	NaN	NaN	NaN	0.638298	
std	278.736281	NaN	NaN	NaN	0.485688	
min	101.000000	NaN	NaN	NaN	0.000000	
25%	222.000000	NaN	NaN	NaN	0.000000	
50%	444.000000	NaN	NaN	NaN	1.000000	
75%	685.250000	NaN	NaN	NaN	1.000000	
max	999.000000	NaN	NaN	NaN	1.000000	

```
# lister toutes les colonnes
dataset.columns

Index(['nom', 'prenom', 'sexe', 'date_naissance', 'lieu_naissance',
      'adresse',
      'commune', 'pays', 'nir', 'organisme', 'employeur', 'bin',
      'nic',
      'date_debut', 'date_fin', 'regime', 'identifie'],
      dtype='object')
```

3. Gestion des valeurs manquantes

```
# vérifier si il existe des valeurs manquantes dans le jeu de données
dataset.isnull().values.any()
```

```
True
```

```
# le nombre de valeurs manquantes dans le jeu de données
dataset.isnull().values.sum()
```

```
199
```

```
# lister les valeurs manquantes
dataset.isna().sum()
```

```
nom          15
prenom       17
sexe         0
date_naissance 13
lieu_naissance 13
adresse       13
commune       11
pays          0
nir          17
organisme     11
employeur     14
bin           11
nic           11
date_debut    12
date_fin      33
regime        8
identifie     0
dtype: int64
```

```
# lister les colonnes contenant des valeurs manquantes
dataset.isna().columns
```

```
Index(['nom', 'prenom', 'sexe', 'date_naissance', 'lieu_naissance',
      'adresse',
```

```

        'commune', 'pays', 'nir', 'organisme', 'employeur', 'bin',
        'nic',
        'date_debut', 'date_fin', 'regime', 'identifie'],
        dtype='object')

# la taille des colonnes contenant des valeurs manquantes
len(dataset.isna().columns)

17

```

4. Nettoyage des données

```

# Remplacer les valeurs incohérentes
dataset = dataset.replace(["", " ", "NULL"], np.nan)

# Conversion types
dataset["date_naissance"] = pd.to_datetime(dataset["date_naissance"],
errors="coerce")
dataset["date_debut"] = pd.to_datetime(dataset["date_debut"],
errors="coerce")
dataset["date_fin"] = pd.to_datetime(dataset["date_fin"],
errors="coerce")

dataset["bin"] = pd.to_numeric(dataset["bin"], errors="coerce")
dataset["nic"] = pd.to_numeric(dataset["nic"], errors="coerce")

# Feature Engineering
dataset["nir_present"] = dataset["nir"].notna().astype(int)
dataset["nom_present"] = dataset["nom"].notna().astype(int)
dataset["prenom_present"] = dataset["prenom"].notna().astype(int)

# Complétude
dataset["nb_champs_renseignes"] = dataset.notna().sum(axis=1)
dataset["taux_complétude"] = dataset.notna().mean(axis=1)

# Age
dataset["age"] = 2026 - dataset["date_naissance"].dt.year

# Durée activité
dataset["duree_activite"] = (dataset["date_fin"] -
dataset["date_debut"]).dt.days

dataset.head(10)

```

	nom	prenom	sexe	date_naissance	lieu_naissance	\
0	Dupont	Alice	F	1985-03-12	Tours	
1	Martin	Bob	M	1978-11-02	Paris	
2	Durand	Charlie	M	1990-07-25	Lyon	
3	Lefevre	Emma	F	1982-01-17	Bordeaux	
4	Moreau	David	M	1975-09-09	Lille	

5	NaN	NaN	M	1988-06-10	Marseille
6	Dubois	NaN	F	NaT	Nice
7	NaN	NaN	F	NaT	NaN
8	Petit	Lucas	M	1995-04-04	Nantes
9	Robert	Chloe	F	1989-12-30	Strasbourg

	adresse	commune	pays	nir
organisme ... \				
0	12 rue Victor Hugo	Tours	France	2.850312e+14
CPAM ...				
1	45 avenue République	Paris	France	1.781103e+14
MSA ...				
2	8 rue Nationale	Lyon	France	1.900725e+14
CPAM ...				
3	22 rue Sainte Catherine	Bordeaux	France	2.820117e+14
RSI ...				
4	3 rue Faidherbe	Lille	France	1.750909e+14
CPAM ...				
5	NaN	Marseille	France	NaN
CPAM ...				
6	15 rue Massena	Nice	France	NaN
MSA ...				
7	NaN	NaN	France	NaN
NaN ...				
8	9 rue Crébillon	Nantes	France	1.950404e+14
CPAM ...				
9	5 rue des Grandes Arcades	Strasbourg	France	2.891230e+14
RSI ...				

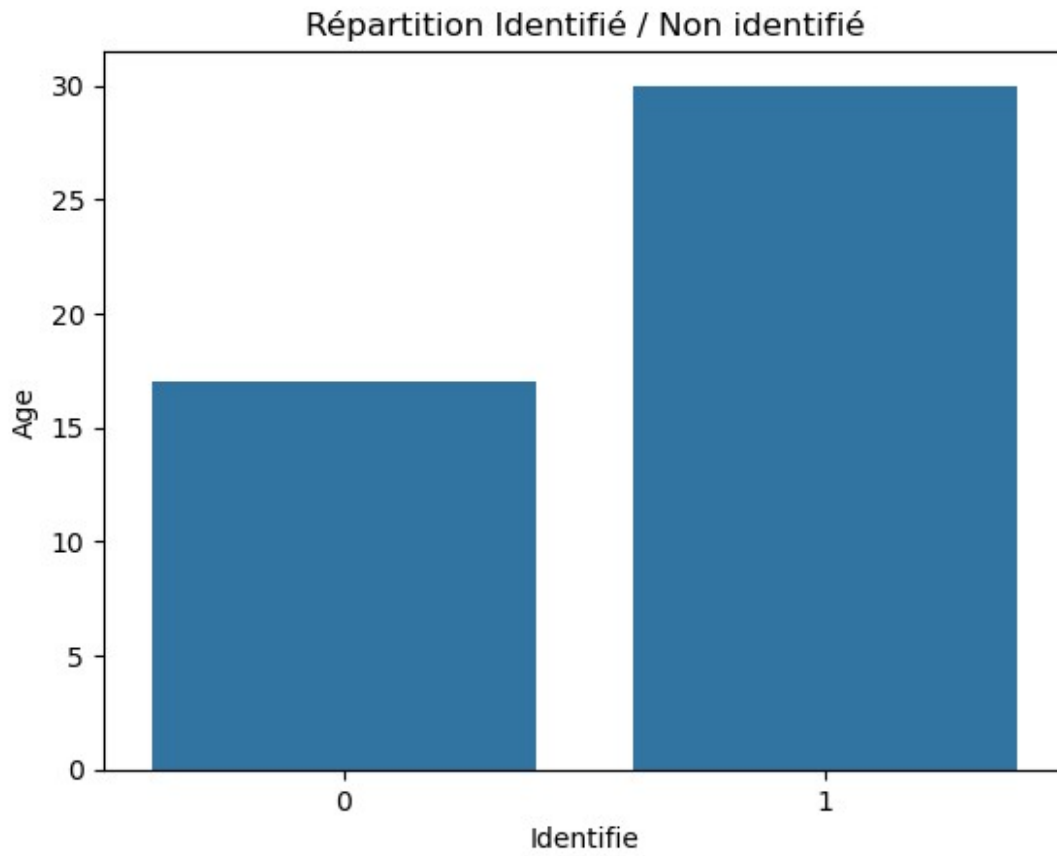
	date_fin	regime	identifie	nir_present	nom_present
prenom_present \					
0	2015-06-30	General	1	1	1
1					
1	2018-12-31	Agricole	1	1	1
1					
2	2020-10-15	General	1	1	1
1					
3	NaT	Independant	1	1	1
1					
4	2010-12-31	General	1	1	1
1					
5	NaT	General	0	0	0
0					
6	NaT	Agricole	0	0	1
0					
7	NaT	NaN	0	0	0
0					
8	NaT	General	1	1	1
1					

9	NaT	Independant	1	1	1
1					
	nb_champs_renseignes	taux_completude	age	duree_activite	
0	20	1.000000	41.0	2006.0	
1	20	1.000000	48.0	5053.0	
2	20	1.000000	36.0	1628.0	
3	19	0.952381	44.0	NaN	
4	20	1.000000	51.0	4017.0	
5	14	0.714286	38.0	NaN	
6	14	0.714286	NaN	NaN	
7	6	0.333333	NaN	NaN	
8	19	0.952381	31.0	NaN	
9	19	0.952381	37.0	NaN	

[10 rows x 24 columns]

5. Visualisation

```
# Distribution cible
sns.countplot(x="identifie", data=dataset)
plt.ylabel("Age")
plt.xlabel("Identifie")
plt.title("Répartition Identifié / Non identifié")
plt.show()
```

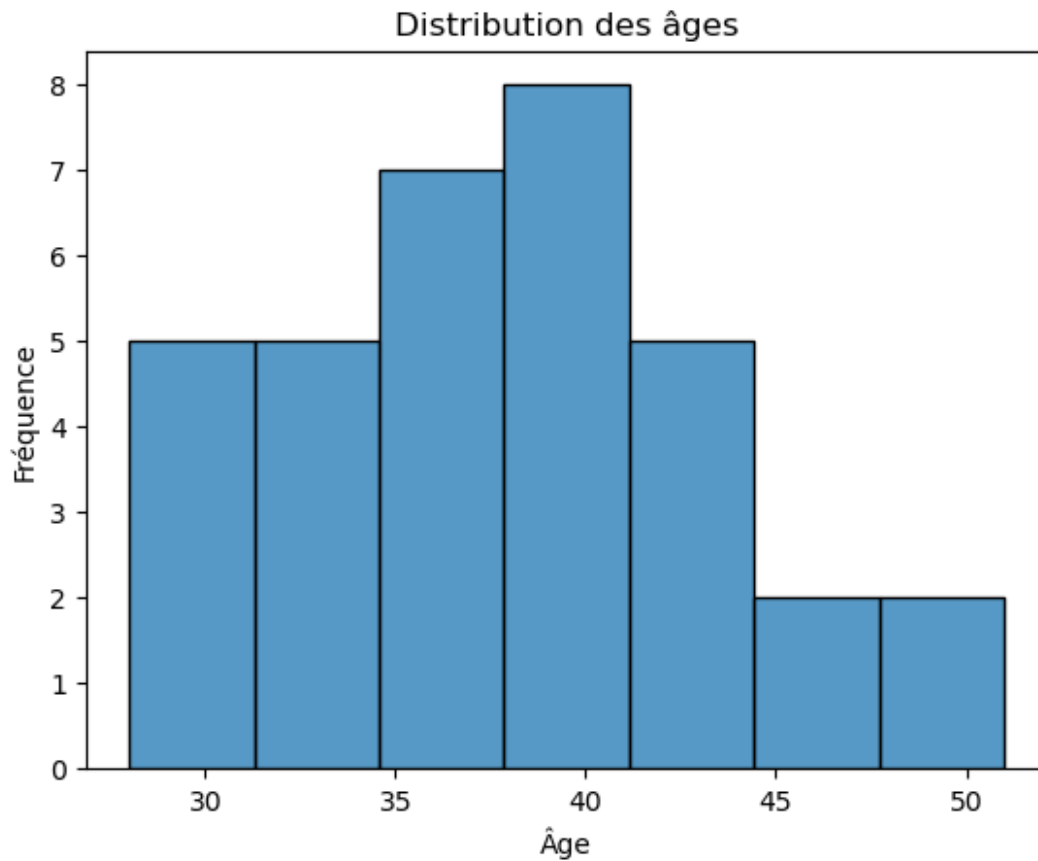



```
# Distribution de la cible
dataset["identifie"].value_counts()

identifie
1      30
0      17
Name: count, dtype: int64
```

Il y a 30 personnes identifiées et 17 personnes non identifiées dans le jeu de données

```
# Distribution Age
sns.histplot(x="age", data=dataset)
plt.ylabel("Fréquence")
plt.xlabel("Âge")
plt.title("Distribution des âges")
plt.show()
```



```
print("La mode est :", dataset['age'].mode()[0])
```

La mode est : 35.0

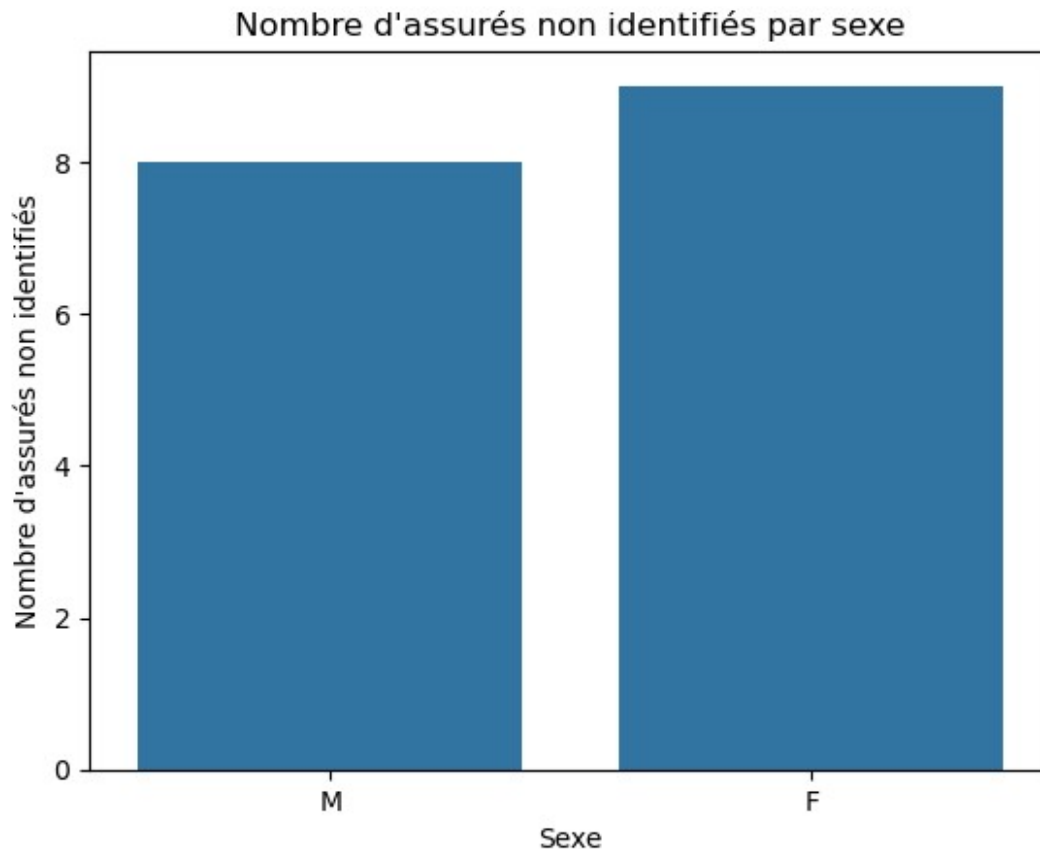
l'âge le plus fréquent dans ce jeu de données est 35 ans

```
# Filtrer les non identifiés
df_non_identifiés = dataset[dataset["identifie"] == 0]

# Plot
sns.countplot(x="sexe", data=df_non_identifiés)

plt.ylabel("Nombre d'assurés non identifiés")
plt.xlabel("Sexe")
plt.title("Nombre d'assurés non identifiés par sexe")

plt.show()
```



```
df_non_identifies["sexe"].value_counts()
```

```
sexe
F      9
M      8
Name: count, dtype: int64
```

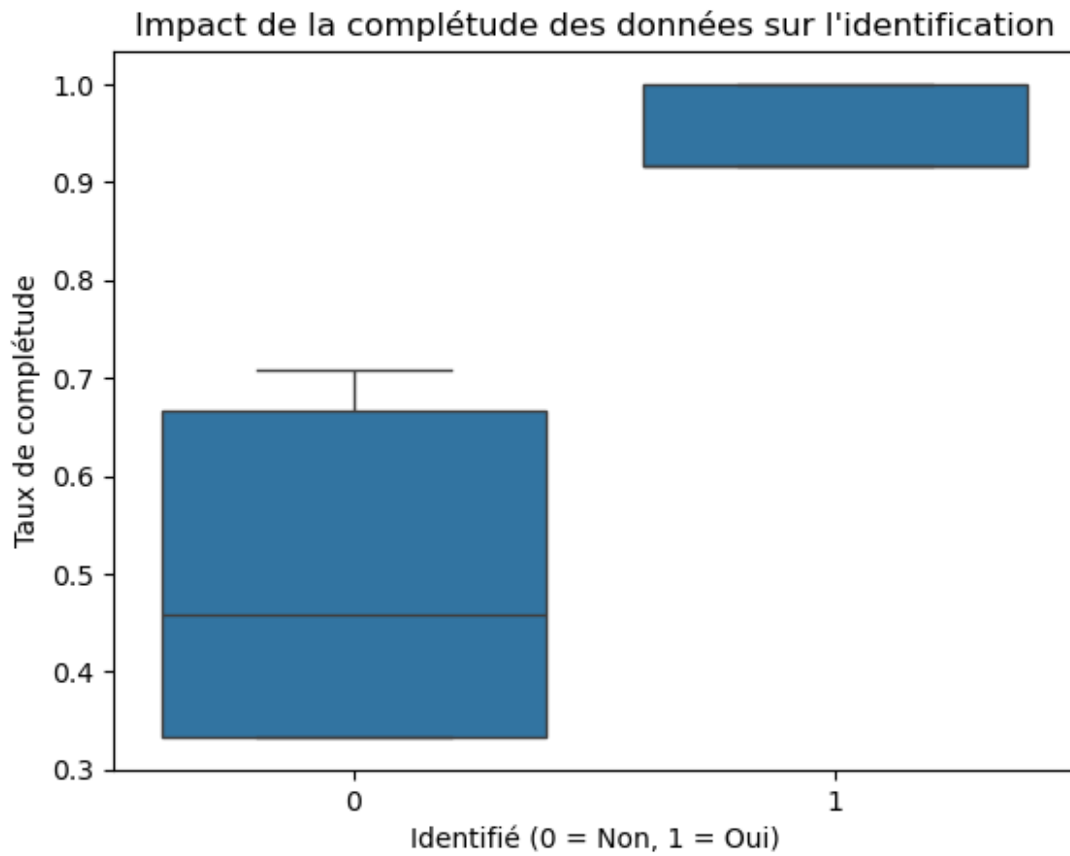
Il y a 9 femmes et 8 hommes non identifiés

```
# taux de complétude vs identification
dataset["taux_complétude"] = dataset.notna().mean(axis=1)

sns.boxplot(x="identifie", y="taux_complétude", data=dataset)

plt.xlabel("Identifié (0 = Non, 1 = Oui)")
plt.ylabel("Taux de complétude")
plt.title("Impact de la complétude des données sur l'identification")

plt.show()
```



On voit clairement que les assurés identifiés ont des données beaucoup plus complètes

```
# matrice de corrélation

# supprimer la variable cible (identifie)
dataset_2 = dataset.drop(columns="identifie")

# sélectionner les variables numériques
dataset_2 = dataset_2.select_dtypes(include=['number'])

# vérifier le jeu de données
dataset_2.head()
```

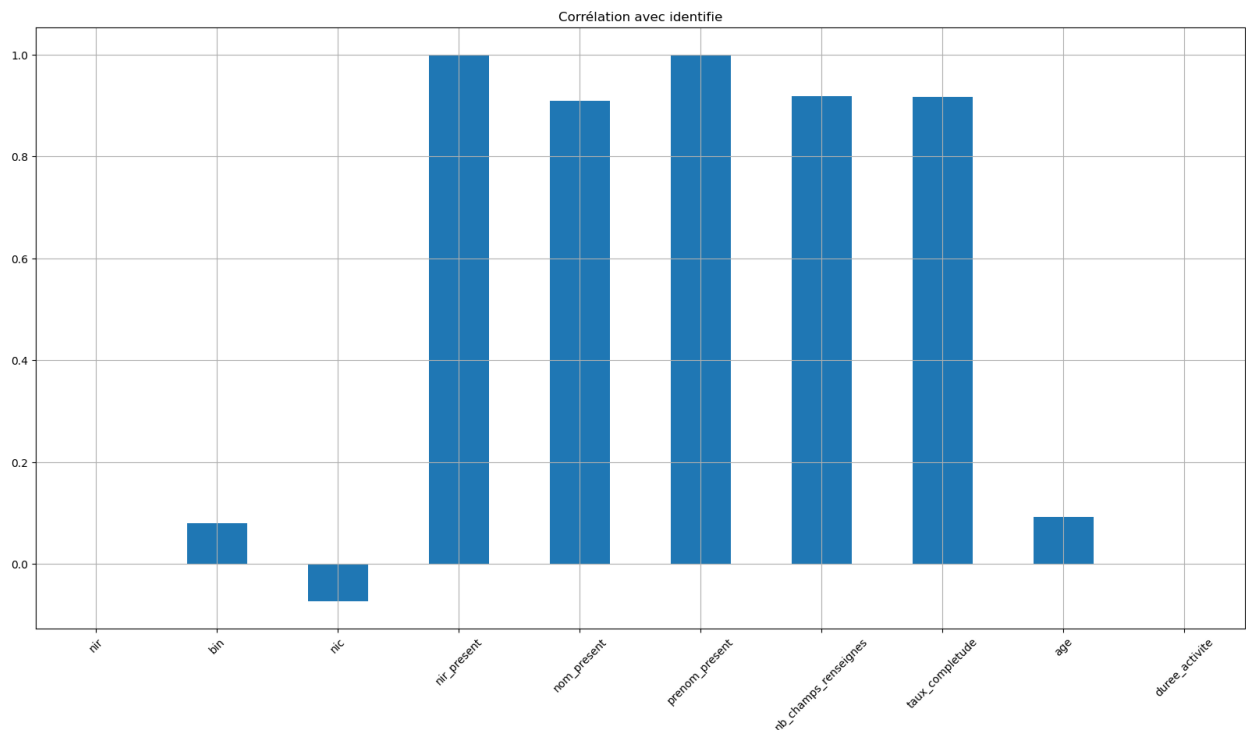
	nir	bin	nic	nir_present	nom_present
prenom_present \					
0	2.850312e+14	12345.0	678.0	1	1
1					
1	1.781103e+14	54321.0	111.0	1	1
1					
2	1.900725e+14	67890.0	222.0	1	1
1					
3	2.820117e+14	22222.0	333.0	1	1
1					
4	1.750909e+14	33333.0	444.0	1	1

1

	nb_champs_reseignes	taux_completude	age	duree_activite
0	20	1.000000	41.0	2006.0
1	20	1.000000	48.0	5053.0
2	20	1.000000	36.0	1628.0
3	19	0.916667	44.0	NaN
4	20	1.000000	51.0	4017.0

```
dataset_2.corrwith(dataset["identifie"]).plot.bar(
    figsize=(20,10), title="Corrélation avec identifie", rot=45,
    grid=True
)
```

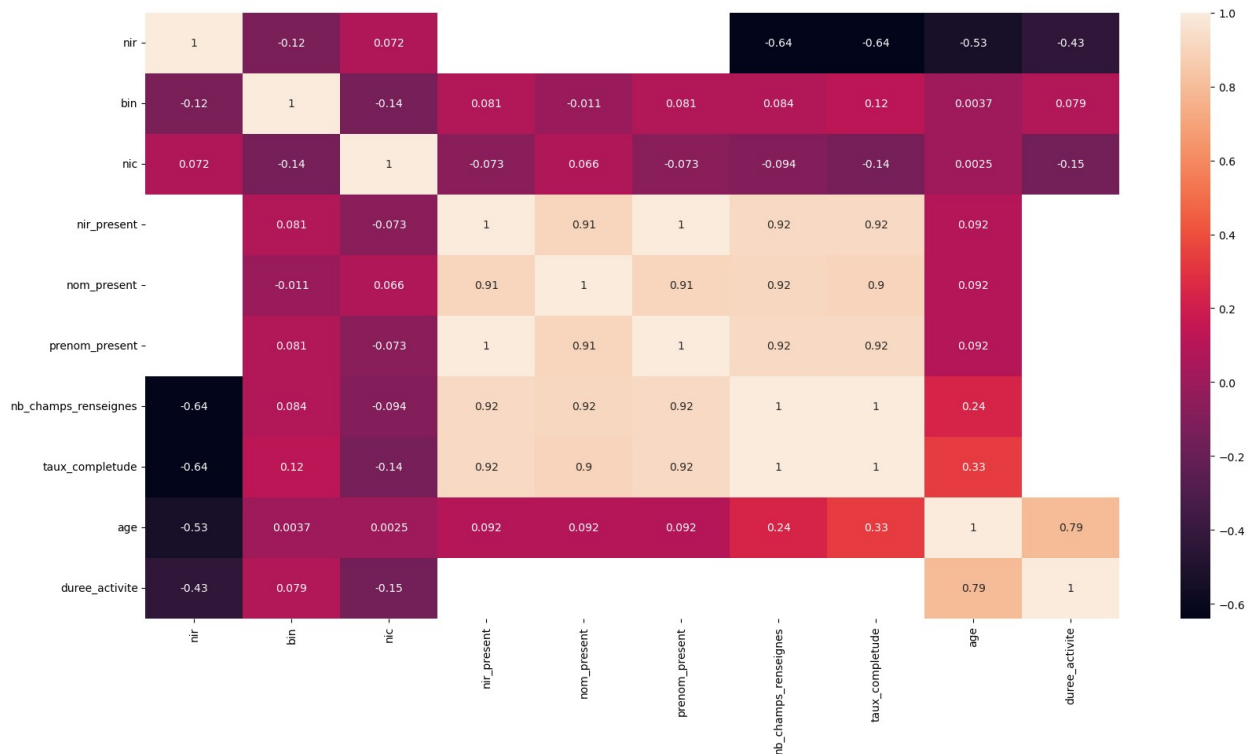
<Axes: title={'center': 'Corrélation avec identifie'}>



Les variables nir_present, nom_present, prenom_present, nbr_champs_reseignes et taux_completude présentent une forte corrélation positive avec la variable cible.

```
# heatmap
corr = dataset_2.corr()
plt.figure(figsize=(20,10))
sns.heatmap(corr, annot=True)
```

<Axes: >



6. Préparation des données

```
# Sélection des features
features = [
    "nir_present",
    "nom_present",
    "prenom_present",
    "nb_champs_renseignes",
    "taux_completude",
    "age",
    "duree_activite"
]

X = dataset[features]
y = dataset["identifie"]

# Remplir NaN
X = X.fillna(0)

X.shape
(47, 7)

y.shape
(47,)
```

7. Train Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

X_train.shape
(37, 7)

X_test.shape
(10, 7)

y_train.shape
(37,)

y_test.shape
(10,)

# Feature scaling (mettre toutes les variables à la même échelle)
from sklearn.preprocessing import StandardScaler

sc = StandardScaler()

X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

X_train
array([[ 0.73598007,  0.69282032,  0.73598007,  0.59258337,
         0.51781586,  0.19073    , -0.655317   ],
       [ 0.73598007,  0.69282032,  0.73598007,  0.59258337,
         0.51781586,  0.93659284, -0.655317   ],
       [-1.35873244,  0.69282032, -1.35873244, -0.33646683, -
         0.47317657, -1.587866   , -0.655317   ],
       [-1.35873244, -1.44337567, -1.35873244, -1.26551702, -
         1.29900359, -1.587866   , -0.655317   ],
       [ 0.73598007,  0.69282032,  0.73598007,  0.77839341,
         0.84814667,  0.76447064,  1.24790927],
       [-1.35873244, -1.44337567, -1.35873244, -0.52227687, -
         0.47317657,  0.42022626, -0.655317   ],
```

[0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
1.05134097, 1.4656878],
[0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
0.0186078 , -0.655317],
[-1.35873244, -1.44337567, -1.35873244, -1.82294714, -
1.79449981,
-1.587866 , -0.655317],
[0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
0.53497439, -0.655317],
[0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
0.30547813, -0.655317],
[-1.35873244, -1.44337567, -1.35873244, -1.26551702, -
1.29900359,
-1.587866 , -0.655317],
[-1.35873244, -1.44337567, -1.35873244, -0.52227687, -
0.47317657,
0.36285219, -0.655317],
[0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
0.87921877, 1.59721741],
[0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
0.76447064, 0.78647752],
[0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
0.30547813, -0.655317],
[0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
0.47760032, 1.16237908],
[-1.35873244, -1.44337567, -1.35873244, -0.33646683, -
0.30801116,
0.59234845, -0.655317],
[0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
0.87921877, 1.53253072],
[-1.35873244, -1.44337567, -1.35873244, -1.82294714, -
1.79449981,
-1.587866 , -0.655317],
[0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
1.1660891 , 2.97648146],
[0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
0.19073 , -0.655317],
[0.73598007, 0.69282032, 0.73598007, 0.59258337,


```

0.51781586,
    0.42022626, -0.655317 ],
    [ 0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
    0.47760032, 0.51479341],
    [ 0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
    0.07598187, -0.655317 ],
    [ 0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
    0.59234845, 1.09409868],
    [-1.35873244, -1.44337567, -1.35873244, -1.82294714, -
1.79449981,
    -1.587866 , -0.655317 ],
    [-1.35873244, -1.44337567, -1.35873244, -1.26551702, -
1.29900359,
    -1.587866 , -0.655317 ],
    [ 0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
    1.10871503, -0.655317 ],
    [-1.35873244, -1.44337567, -1.35873244, -1.82294714, -
1.79449981,
    -1.587866 , -0.655317 ],
    [-1.35873244, -1.44337567, -1.35873244, -1.82294714, -
1.79449981,
    -1.587866 , -0.655317 ],
    [ 0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
    0.70709658, 1.22634704],
    [-1.35873244, -1.44337567, -1.35873244, -1.82294714, -
1.79449981,
    -1.587866 , -0.655317 ],
    [ 0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
    0.64972252, 1.42256334],
    [ 0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
    0.64972252, 1.35643916],
    [ 0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
    0.82184471, -0.655317 ],
    [ 0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
    0.53497439, -0.655317 ]])

```

X_test

```

array([[ 0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
    0.13335593, -0.655317 ],

```

```

[ 0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
0.42022626, -0.655317 ],
[-1.35873244, -1.44337567, -1.35873244, -1.82294714, -
1.79449981,
-1.587866 , -0.655317 ],
[-1.35873244, -1.44337567, -1.35873244, -1.82294714, -
1.79449981,
-1.587866 , -0.655317 ],
[-1.35873244, -1.44337567, -1.35873244, -0.33646683, -
0.30801116,
0.64972252, -0.655317 ],
[ 0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
0.93659284, 1.42256334],
[ 0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
0.36285219, -0.655317 ],
[ 0.73598007, 0.69282032, 0.73598007, 0.59258337,
0.51781586,
0.24810406, -0.655317 ],
[ 0.73598007, 0.69282032, 0.73598007, 0.77839341,
0.84814667,
1.33821129, 2.23186574],
[-1.35873244, 0.69282032, -1.35873244, -0.33646683, -
0.47317657,
-1.587866 , -0.655317 ]])

```

8. Création du modèle

Régression Logistique

```

from sklearn.linear_model import LogisticRegression

# créer le modèle
classif_l_r = LogisticRegression(random_state=0)

# entraîner le modèle
classif_l_r.fit(X_train, y_train)

LogisticRegression(random_state=0)

# faire des predictions / le jeu de test
y_pred = classif_l_r.predict(X_test)

# analyser les performances du model "Logistic Regression"
from sklearn.metrics import confusion_matrix, accuracy_score,
precision_score , f1_score, recall_score

```

```
# définir les métriques
acc = accuracy_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
prec = precision_score(y_test, y_pred)
rec = recall_score(y_test, y_pred)

# define a dataframe
results = pd.DataFrame([["Logistic Regression", acc, f1, prec, rec]],
                        columns=["Model", "Accuracy", "F1-Score", "Precision", "Recall"])

results
```

	Model	Accuracy	F1-Score	Precision	Recall
0	Logistic Regression	1.0	1.0	1.0	1.0

Le modèle "régression logistique" fait 100% de bonnes prédictions

```
# la matrice de confusion
cm = confusion_matrix(y_test, y_pred)
print(cm)

[[4 0]
 [0 6]]
```

Interprétation : 4 vrais négatifs (bien classés) 6 vrais positifs (bien classés) 0 erreur

9. Validation croisée

```
from sklearn.model_selection import cross_val_score

# on utilise le score de la validation croisée pour évaluer les
performances du modèle
from sklearn.model_selection import cross_val_score

accuracies = cross_val_score(estimator=classif_l_r, X=X_train,
                              y=y_train, cv=10)

print("Accuracy: {:.2f} %".format(accuracies.mean() * 100))
print("Standard Deviation: {:.2f} %".format(accuracies.std() * 100))

Accuracy: 100.00 %
Standard Deviation: 0.00 %
```

Étant donné que l'accuracy obtenue avec la régression logistique est de 100 %, probablement en raison de la petite taille du jeu de données (47 observations et 15 caractéristiques), je ne vais pas créer plusieurs modèles d'IA et les comparer pour choisir le meilleur.

Cependant, je vais optimiser le modèle existant (`classifier_lr`) avec `RandomizedSearchCV` afin de trouver les meilleurs hyperparamètres, uniquement dans un but de démonstration.

10. Optimisation avec RandomizedSearchCV

```
from sklearn.model_selection import RandomizedSearchCV

# définir la grille d'hyperparamètres pour RandomizedSearchCV
param_grid = {
    "C": [0.01, 0.1, 1, 10, 100],
    "penalty": ["l1", "l2", "elasticnet", "none"],
    "solver": ["liblinear", "saga"] # 'saga' nécessaire si
    penalty='elasticnet'
}

# créer le RandomizedSearchCV
search = RandomizedSearchCV(
    classifier_lr,
    param_grid,
    n_iter=10, # nombre d'itérations aléatoires
    cv=3,      # cross-validation 3-fold
    random_state=42
)

# entraîner le modèle
search.fit(X_train, y_train)

# Meilleur modèle
best_model = search.best_estimator_

print("Meilleur score :", search.best_score_)
print("Meilleurs hyperparamètres :", search.best_params_)

Meilleur score : 1.0
Meilleurs hyperparamètres : {'solver': 'saga', 'penalty': 'l2', 'C': 1}
```

11. Évaluation finale

```
# faire des predictions avec le meilleur modèle
y_pred = best_model.predict(X_test)

# définir les métriques
acc = accuracy_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
prec = precision_score(y_test, y_pred)
```

```

rec = recall_score(y_test, y_pred)

final_model_results = pd.DataFrame([["Final Logistic Regression", acc,
f1, prec, rec]],
                                columns=["Model", "Accuracy", "F1-Score",
"Precision", "Recall"])

result = pd.concat([results, final_model_results], ignore_index=True)

result

```

	Model	Accuracy	F1-Score	Precision	Recall
0	Logistic Regression	1.0	1.0	1.0	1.0
1	Final Logistic Regression	1.0	1.0	1.0	1.0

12. Tester sur une nouvelle observation

```

new_data = pd.DataFrame([{
    "nir_present": 1,
    "nom_present": 1,
    "prenom_present": 1,
    "nb_champs_renseignes": 15,
    "taux_completude": 0.9,
    "age": 35,
    "duree_activite": 2000
}])

prediction = best_model.predict(new_data)

print("Résultat :", prediction)

Résultat : [1]

```

Interprétation : La nouvelle observation testée est prédite comme "identifiée" par le modèle

```

# sauvegarder le meilleur modèle
import joblib

joblib.dump(best_model, "model.pkl")

['model.pkl']

```